Meet 0

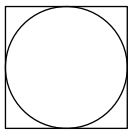
October 19, 2026

Geometry

The answer key is below.

p	Answer	p	Answer
1	86	4	16
2	80	5	10
3	60	6	42

1. Alice and Bob are running a race around a square track with side length 100 meters. Alice runs around the full length of the square, but Bob cuts corners by instead running around the square's inscribed circle. How much further does Alice run than Bob, rounded to the nearest integer?

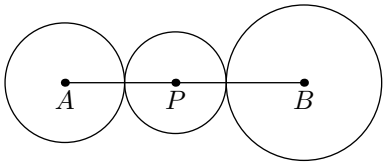


Proposed by: Michael Luo

Answer: $\boxed{86}$

Solution: The perimeter of the square is $4 \cdot 100 = 400$, while the perimeter of the circle is 100π, which is 314 to the nearest integer. So the answer is $400 - 314 = \boxed{86}$.

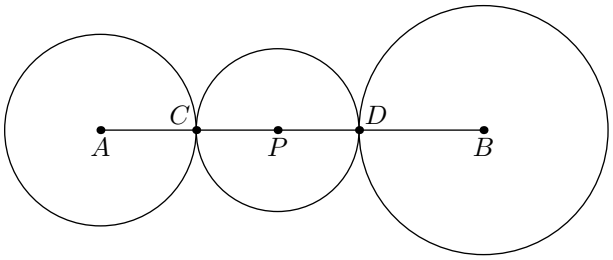
2. Circles X and Y with centers A and B have radii 20 and 26, respectively. Let Z be the circle externally tangent to X and Y with center P on $\overline{AB}$. Suppose $\overline{AP} = 37$. Compute the length of $\overline{AB}$.



Proposed by: Ahmed Ilyasov

Answer: $\boxed{80}$

Solution:



Let C be the point of intersection between circles X and Z. Note that $\overline{AC} = 20$ and $\overline{AP} = 37$, so $\overline{PC} = 17$. Let D be the point of intersection between circles Y and Z. Note that

$$\overline{PB} = \overline{PD} + \overline{BD} = \overline{PC} + \overline{BD} = 17 + 26 = 43.$$

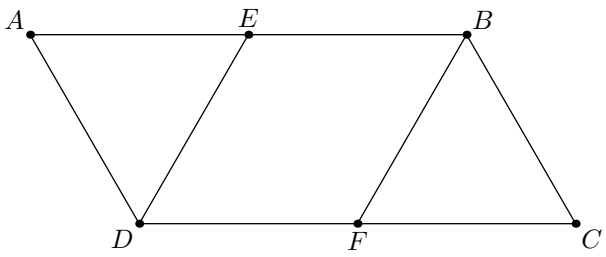
Thus, $\overline{AB} = \overline{AP} + \overline{PB} = 37 + 43 = \boxed{80}$.

3. Parallelogram $ABCD$ satisfies $CD > BC$. Point E lies on side AB so that $AD = DE$, and point F lies on side CD so that $BF = FC$. If $DE \parallel BF$, what is $\angle BAD$, in degrees?

Proposed by: Jeff Zhou

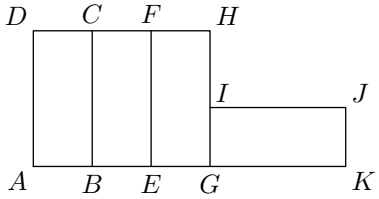
Answer: $\boxed{60}$

Solution:



Let $\angle BAD = \theta$; by the given isosceles triangles and because $ABCD$ is a parallelogram, we also have $\angle AED = \theta = \angle BCF = \angle FBC$. Thus, $\angle BFC = 180^\circ - 2\theta$, so $\angle DFB = 2\theta$. Also, $\angle ADE = 180^\circ - 2\theta$ and $\angle ADC = 180^\circ - \theta$, so $\angle EDF = \angle ADF - \angle ADC = \theta$. Now, because $DE \parallel BF$, we have $180^\circ = \angle EDF + \angle DFB = \theta + 2\theta = 3\theta$, so $\theta = \boxed{60^\circ}$.

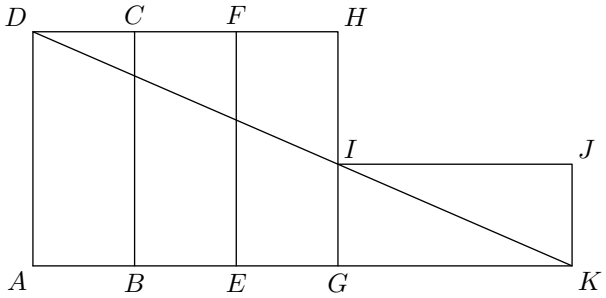
4. In the figure below, $ABCD$, $BEFC$, $EGHF$, and $IGKJ$ are congruent rectangles, so that I lies on segment GH. Points K, I, and D are collinear. The ratio $\frac{AD}{AB}$ is $\frac{a+\sqrt{b}}{c}$ for some positive integers a, b, and c, where b is not divisible by the square of any prime. What is $a+b+c$?



Proposed by: Jeff Zhou

Answer: $\boxed{16}$

Solution:



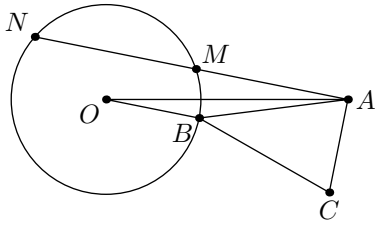
Without loss of generality, let $AB = 1$, and let $AD = x$. Then, observe that since $\angle HID = \angle GIK$ and $\angle IHD = 90^\circ = \angle IGK$, we have $\triangle HID \sim \triangle GIK$ by AA similarity. We have $GK = x$, $GI = 1$, $HI = x - 1$, and $HD = 3$, so

$$\frac{GI}{GK} = \frac{1}{x} = \frac{x-1}{3} = \frac{HI}{HD},$$

meaning $x^2 - x - 3 = 0$. By the quadratic formula, the positive solution to this is $x = \frac{1+\sqrt{13}}{2}$, which is also the ratio of AD to AB. Thus the answer is $1 + 13 + 2 = \boxed{16}$.

Remark (Author's comments)**.** This problem came to me in a dream. Basically uhhh people (including Michael) were looking and critiquing my scratch paper over Zoom and then I randomly drew this diagram but with 3 rectangles instead of 4. Unfortunately 3 rectangles is too easy so I went with 4 instead.

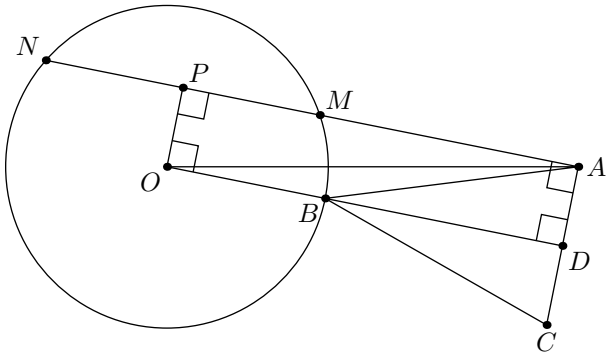
5. Let ω be a circle with radius 2 centered at O, and let A be a point outside ω. A line ℓ passing through A intersects ω at M and N, where $AM < AN$. Another line ℓ_1 is constructed parallel to ℓ through O. Suppose ℓ_1 intersects ω at a point B on a different side of AO as M. Let C be the reflection of A across ℓ_1. The area of $\triangle ABO$ is 1, and the area of $\triangle ABC$ is 3. Compute $AM + AN$.



Proposed by: Jeff Zhou

Answer: $\boxed{10}$

Solution:



Let P be the midpoint of MN and D be the midpoint of AC; the latter is also the foot of the altitude from A to ℓ_1. We wish to compute $AM + AN = 2AP$. Observe that $\angle OPA = \angle ADO = 90^\circ$, so since $AP \parallel DO$, $APOD$ is a rectangle.

Now, $1 = [ABO] = \frac{1}{2}BO \cdot AD = AD$. Thus, $AC = 2AD = 2$, so $3 = [ABC] = \frac{1}{2}BD \cdot AC = BD$. Now, $AP = OD = OB + BD = 2 + 3 = 5$, so $2AP = \boxed{10}$.

6. Let N be the unique positive integer such that

$$\frac{\cot\frac{\pi}{3}\cot\frac{\pi}{7}\cot\frac{\pi}{N}}{\cot\frac{\pi}{3}+\cot\frac{\pi}{7}+\cot\frac{\pi}{N}} = 1.$$

Compute N.

Proposed by: Ahmed Ilyasov

Answer: $\boxed{42}$

Solution: The idea is to solve for $\cot\frac{\pi}{N}$. We have

$$\begin{aligned}\cot\frac{\pi}{N} &= \frac{\cot\frac{\pi}{3}+\cot\frac{\pi}{7}}{\cot\frac{\pi}{3}\cot\frac{\pi}{7}-1} \\ &= \frac{\tan\frac{\pi}{3}+\tan\frac{\pi}{7}}{1-\tan\frac{\pi}{3}\tan\frac{\pi}{7}} \\ &= \tan\left(\frac{\pi}{3}+\frac{\pi}{7}\right)\end{aligned}$$

$$\begin{aligned}
&= \tan \frac{10\pi}{21} \\
&= \cot \left(\frac{\pi}{2} - \frac{10\pi}{21} \right) \\
&= \cot \frac{\pi}{42},
\end{aligned}$$

where in the second step we multiplied the numerator and denominator by $\tan \frac{\pi}{3} \tan \frac{\pi}{7}$ and used the tangent addition identity in the third. Therefore, the answer is $\boxed{42}$.